

Three designs, one *drug*.

*A 1,000-replicate Monte Carlo on randomization, balance,
and what logistic regression quietly hides.*

SPEAKER	N PER TRIAL	REPLICATES	DESIGNS × MODELS
Sriya Leesha Gourammagari	200	1,000	3 × 2

THE QUESTION

A new drug works.
How we randomize *changes the answer.*

Across three randomization rules and two analysis strategies, which combinations recover the truth and which mislead us?

GROUND TRUTH IN THIS SIMULATION

$$\beta_{\text{drug}} = +1.00 \quad (\text{i.e., OR} \approx 2.72)$$

Everything that follows is measured against this number.

The data-generating process.

$$\begin{aligned} \text{logit } P(Y = 1) = & \\ & -0.30 + +1.00 \cdot \text{Drug} \\ & + +3.00 \cdot \text{Age}_{65+} + -3.00 \cdot \\ & \text{Severe} \\ & + -0.05 \cdot \text{BMI} + -0.03 \cdot \text{SBP}_0 \end{aligned}$$

WHAT THE PARAMETERS SAY

Drug raises log-odds of success by 1.

Age & Severity swamp everything — strong prognostic factors.

BMI & SBP are weak nuisance noise.

PER TRIAL

N = 200 · 1,000 replicates · power sweep N = 50
→ 300

THE CONTESTANTS

Three randomization designs.

Simple

Independent Bernoulli(0.5)

Each patient: a fair coin flip,
independent of everyone else.

Goal: maximum simplicity, no assumptions.

Pocock-Simon

Biased coin $p = 0.75 \rightarrow$ less imbalanced
arm

Minimize current imbalance on Age
and Severity at every assignment.

Goal: covariate balance & precision.

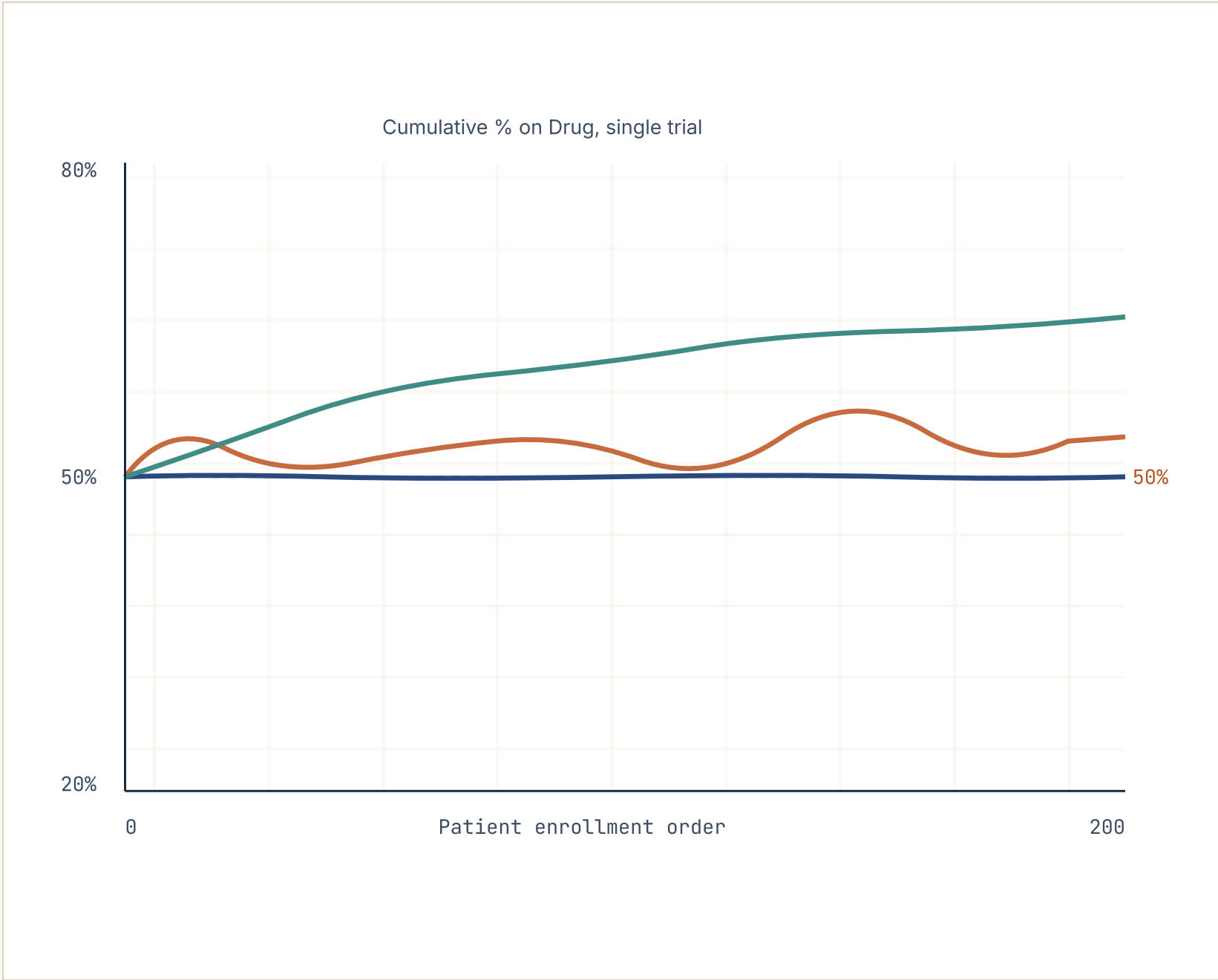
Randomized Play-the-Winner

Urn rule, $\beta = 3$ — success $\rightarrow +3$ same;
failure $\rightarrow +3$ opposite

Urn drifts toward whichever arm is
winning as the trial unfolds.

Goal: ethical allocation, favor winners.

What one trial looks like.



- Pocock locks at 50%.
- Simple wanders.
- RPW drifts up — drug works, urn rewards.

54%

SIMPLE

48.5%

POCOCK

63.5%

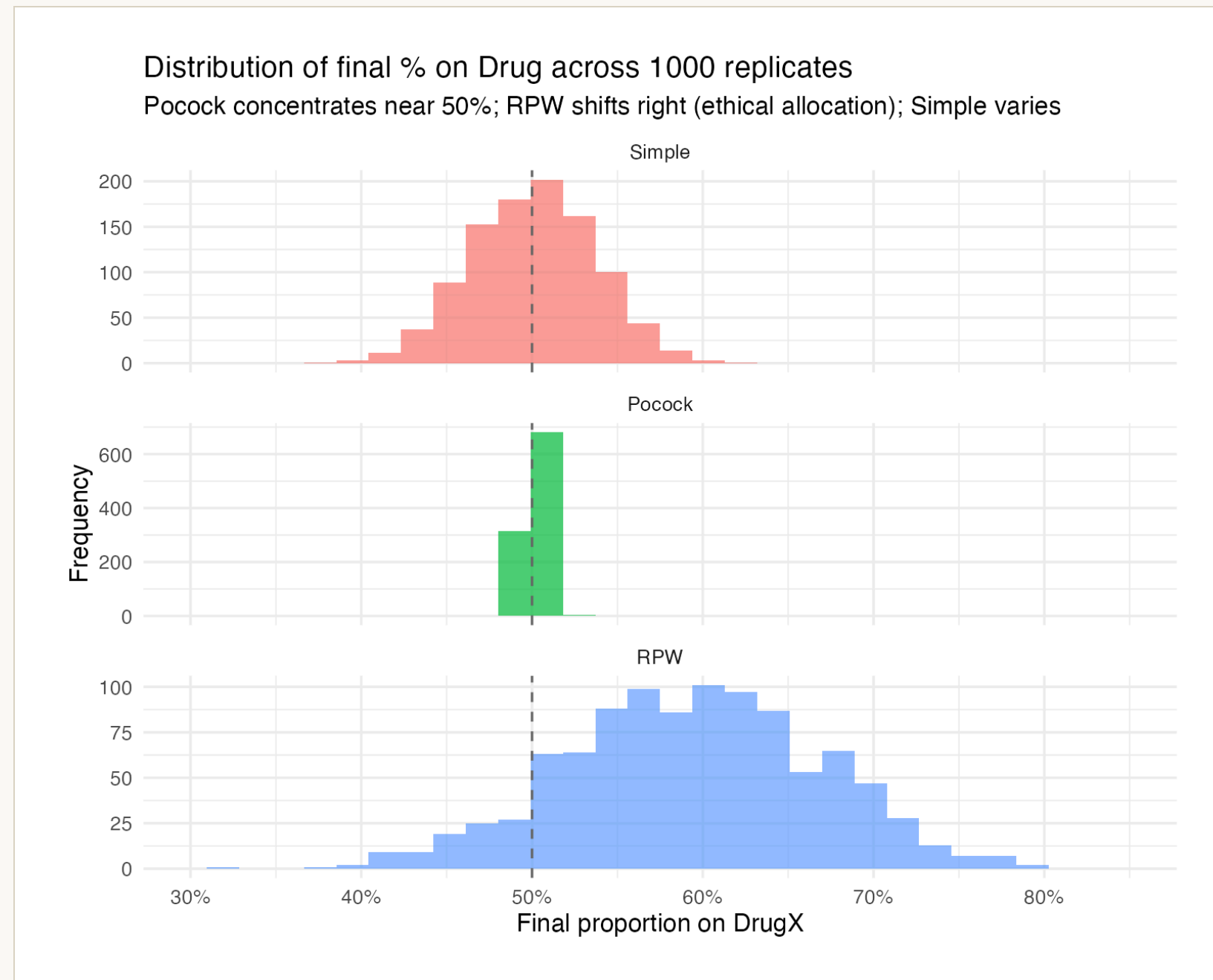
RPW

% on Drug at N = 200, this run.

Each design does what it was built to do, but one trial is an anecdote.

ALLOCATION · 1,000 REPLICATES

The personality of each design.



• SD 0.036

Simple — clean bell around 50%.

• SD 0.006

Pocock — a needle on 50%.

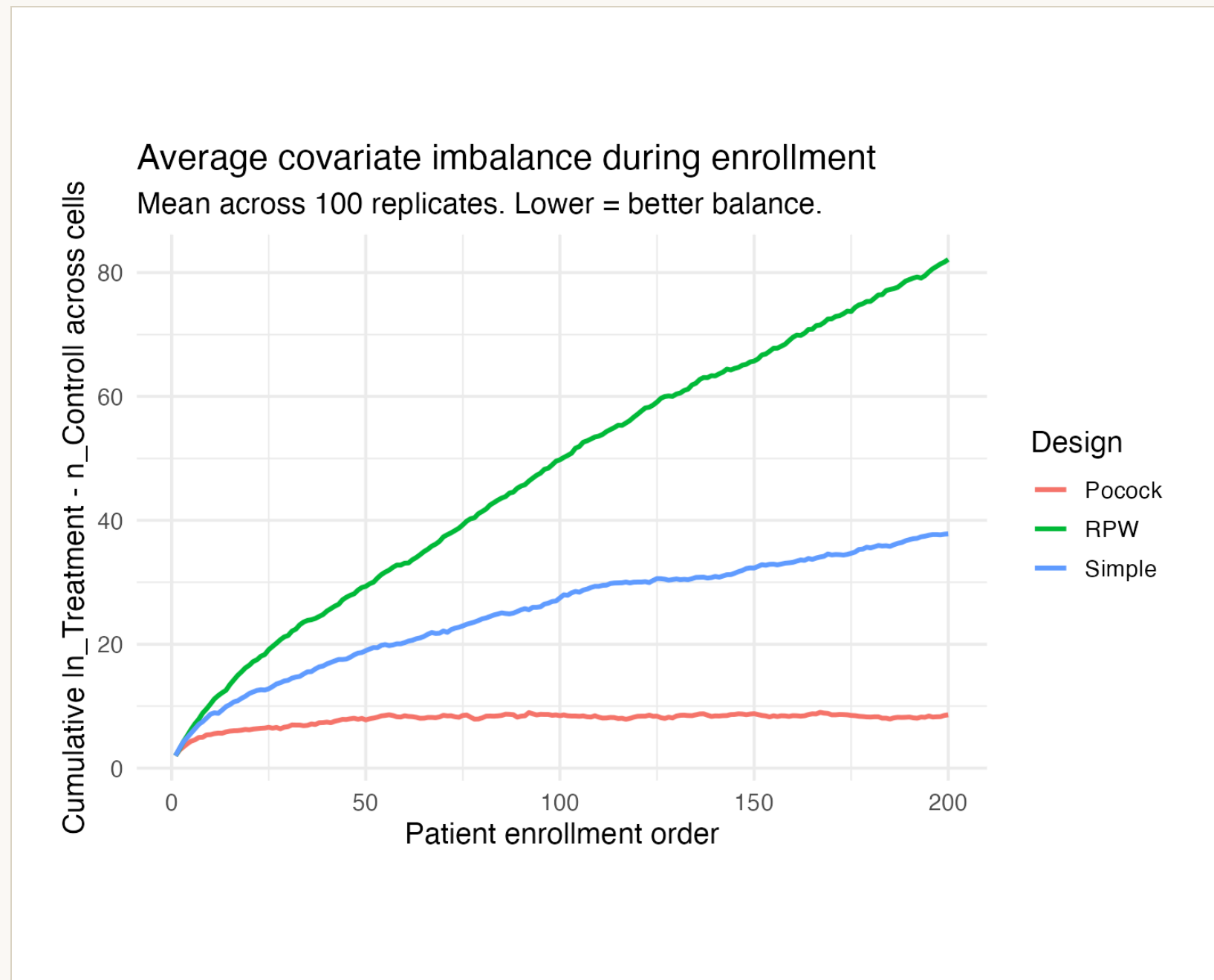
• SD 0.074

RPW — centered $\approx 60\%$, $\sim 12\times$ Pocock's spread.

The shape of the distribution is the personality of the design.

COVARIATE BALANCE DURING ENROLLMENT

Allocation is not the same as balance.



Imbalance summed across Age × Severity cells, mean over 100 replicates.

8.2
POCOCK

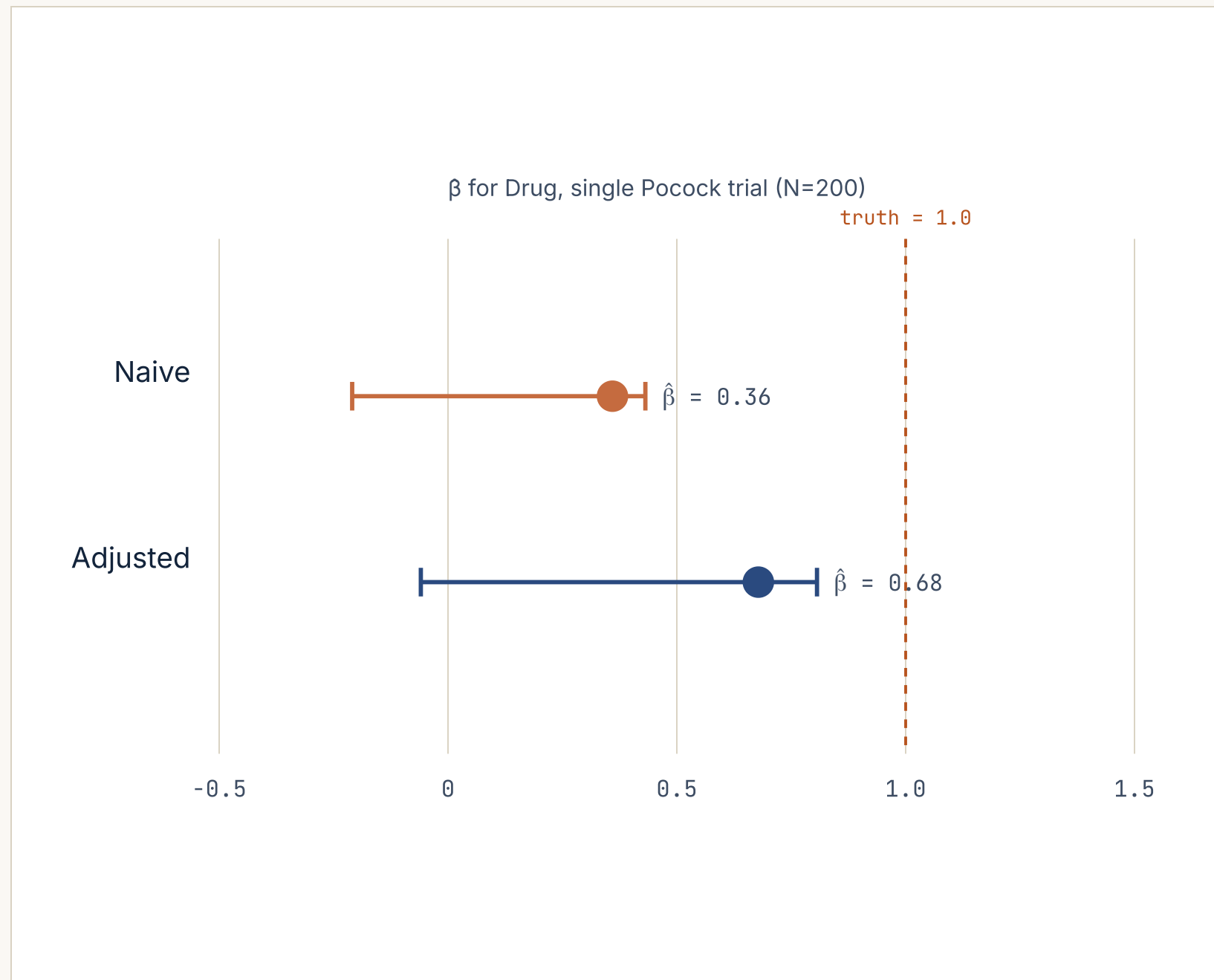
38.8
SIMPLE

86.2
RPW

Mean total categorical imbalance at N = 200.

RPW pays for ethical allocation with the worst covariate balance.

From one trial, a fog.



Both estimates miss truth. Both CIs are wide.

Is this *bias* or *noise*?

From a single trial, it cannot be determined

So I stopped looking at one trial.

METHODOLOGICAL PIVOT

One trial → a thousand.

FROM

A single estimate, with single-trial sampling noise.

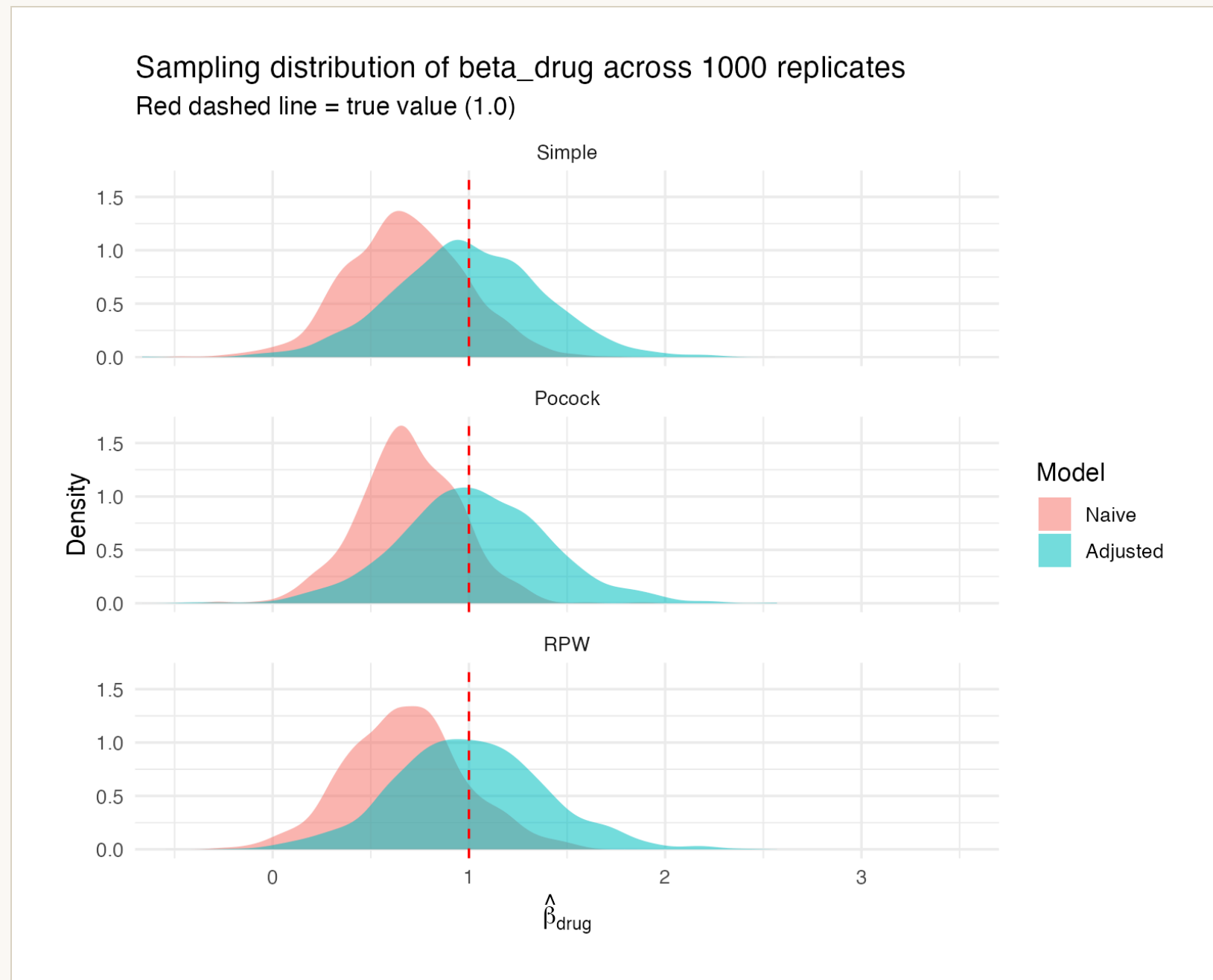
Re-simulate the trial 1,000 times. The mean across replicates is what each design-by-analysis combination does *on average*. The spread is its intrinsic precision.

TO

A sampling distribution, separating *systematic* behavior from *noise*.

SAMPLING DISTRIBUTIONS OF $B_{\text{DRUG}} \cdot 1,000$ REPS

The naive model misses the truth. *In every design.*



≈ 0.68

Naive mean $\hat{\beta}$ — every design.

≈ 1.00

Adjusted mean $\hat{\beta}$ — every design.

A **32% attenuation** from a model, not from a design.

Adjusted recovers $\beta = 1.00$; naive recovers the *marginal* log-OR — mathematically smaller. See next slide.

The design did not cause the bias. The analysis did.

WHY THIS HAPPENS

Non-collapsibility, briefly.

Linear model

$$E[Y \mid X, Z] = \alpha + \beta X + \gamma Z$$

Omit a balanced Z → β unchanged, variance up.
Effect is *collapsible*.

$$\beta = \beta_{\text{marg}} \text{ SAME ANSWER}$$

Logistic model

$$\text{logit } P(Y=1 \mid X, Z) = \alpha + \beta X + \gamma Z$$

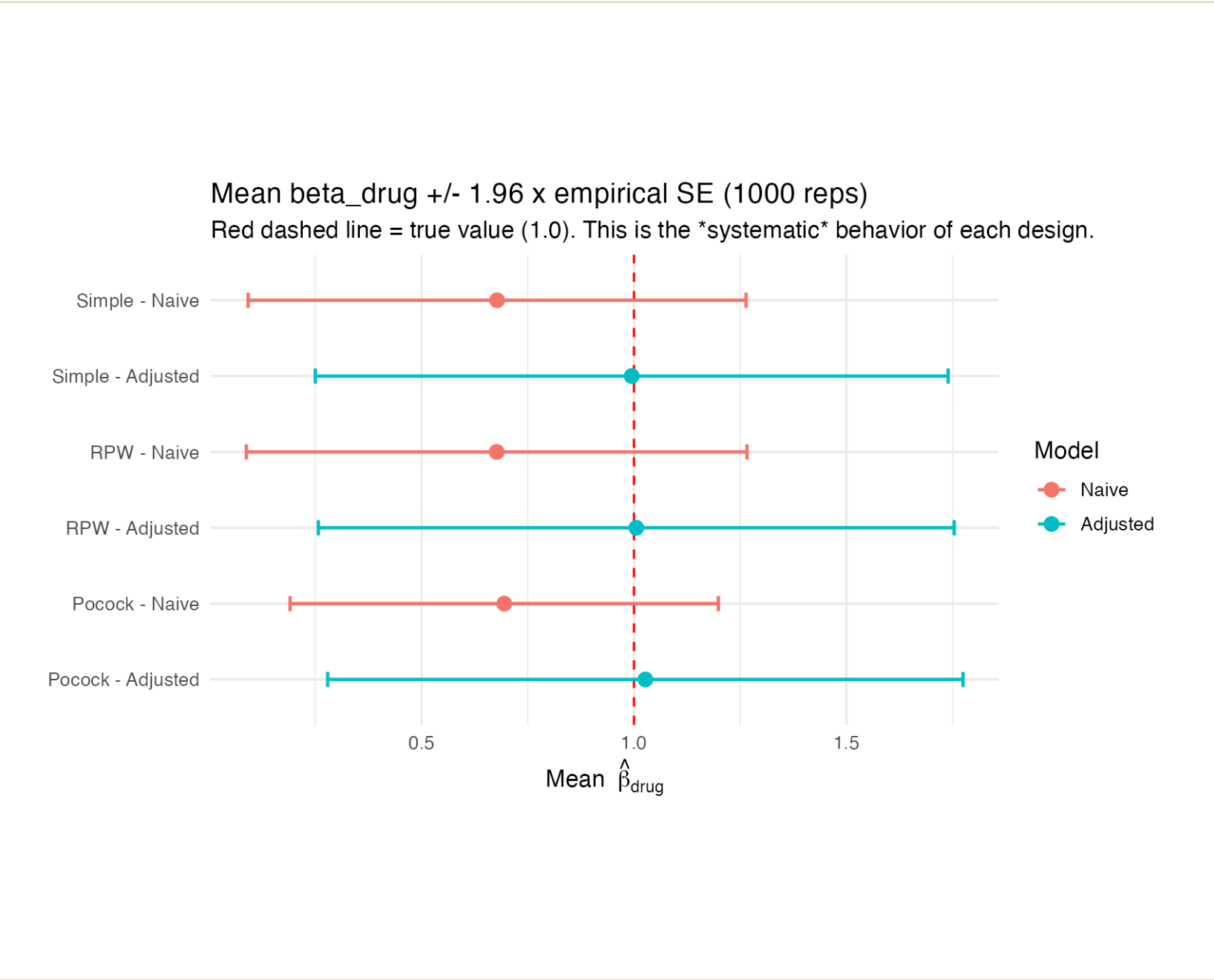
Marginalize over Z → log-OR *shrinks toward 0*.
Effect is *non-collapsible*.

1.00 CONDITIONAL
(TRUTH)

≈0.68 MARGINAL
(NAIVE)

Naive estimates the marginal log-OR; the conditional truth is what we want.

Across designs, the model decides.

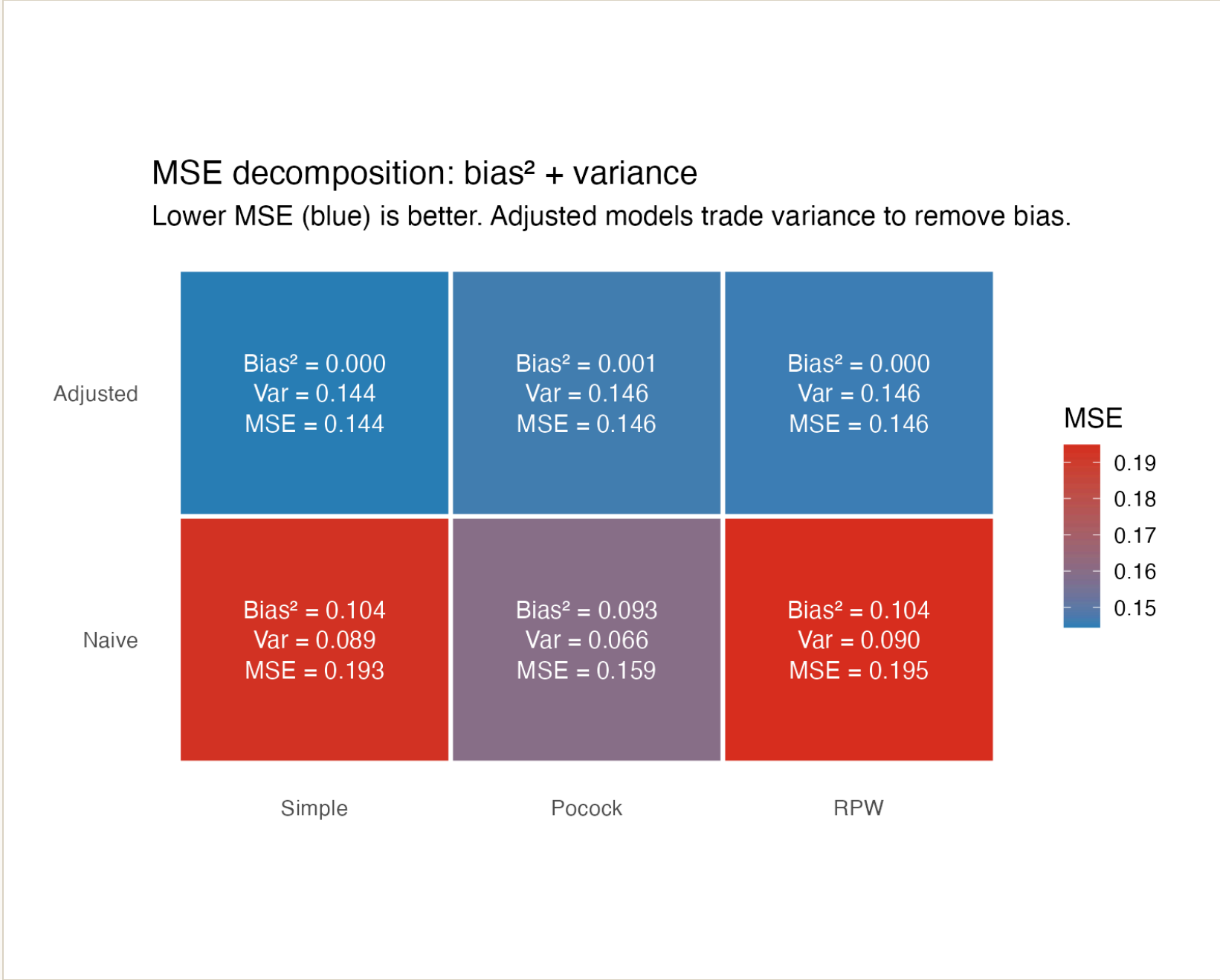


DESIGN × MODEL	MEAN $\hat{\beta}$	EMPSE	COVER
Simple · Naive	0.68	0.299	78.5%
Simple · Adjusted	0.99	0.380	94.2%
Pocock · Naive	0.69	0.257	86.0%
Pocock · Adjusted	1.03	0.381	93.6%
RPW · Naive	0.68	0.301	80.0%
RPW · Adjusted	1.01	0.382	94.2%

Adjustment delivers the truth. Designs are nearly tied at N = 200.

$BIAS^2 + VARIANCE = MSE$

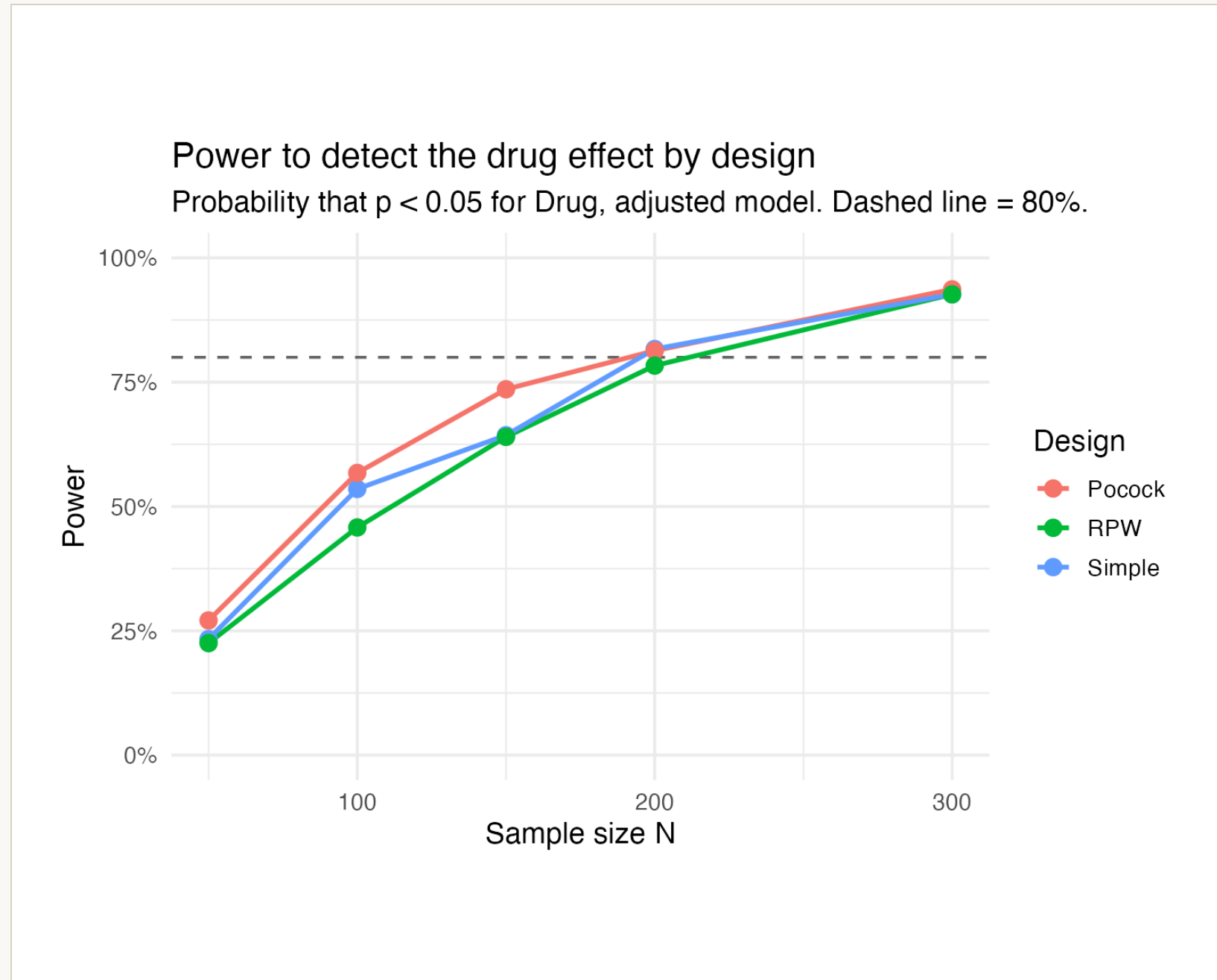
A small variance cost buys zero bias.



CELL	BIAS ²	VAR	MSE
Simple · Naive	0.104	0.089	0.193
Pocock · Naive	0.093	0.066	0.159
RPW · Naive	0.104	0.090	0.195
Simple · Adjusted	0.000	0.144	0.144
Pocock · Adjusted	0.001	0.146	0.146
RPW · Adjusted	0.000	0.146	0.146

Adjustment cuts MSE by ~25%. Design barely moves it.

When does design matter? Under-power.



74%

Pocock at $N = 150$.

64%

Simple / RPW at $N = 150$.

10-point gap — vanishes by $N = 200$.

All three first cross 80% near **$N \approx 200$** — the practical design target.

Design efficiency matters most when you're under-powered.

A WARNING · WHAT IF WE LET THE DATA PICK COVARIATES?

LASSO at λ_{1SE} deletes the drug effect.

METHOD	$\hat{B}_{\text{DRUG}}$	VS TRUTH (1.00)
Full model	0.66	-34%
Stepwise AIC	0.68	-32%
LASSO · λ_{min}	0.49	-51%
LASSO · λ_{1se}	0.04	-96%

The 1-SE rule is a *prediction* heuristic.

Used for *inference*, it can shrink a real effect to zero.

A teaching moment: penalty selected for parsimony, not for unbiased treatment estimation.

Stepwise AIC keeps Drug, Age, Severity — drops both noise covariates.

Choose your penalty for the question you're actually asking.

RPW treats more patients with the working drug.

Ethical win

- **+27 patients** on Drug vs Simple (127 vs 100, single run).
- ~60% allocation when the drug works.
- Real-time response-adaptive — urn updates with every outcome.

Statistical cost

- Worst categorical balance (mean 86.2 vs Pocock's 8.2).
- Largest SD in allocation (0.074 vs 0.006).
- Lowest power below N = 200.

RPW shifts the trade-off toward patients, at a measurable statistical cost.

When to reach for which design.

SITUATION	RECOMMENDATION	WHY
Regulatory Phase III, equal allocation	● Pocock-Simon + Adjusted	Maximum precision; balances on prognostic factors.
Strong known prognostic factors	● Pocock-Simon + Adjusted	Minimization removes random imbalance on Age, Severity.
Pediatric / oncology, large effect	● RPW + Adjusted	Ethical priority; allocates more to the winning arm.
Small pilot / Phase I	● Simple + Adjusted	No assumptions; robust; transparent.
Any of the above with naive analysis	● Avoid	Non-collapsibility shrinks β by ~32%.

Pick the design for who you treat. Adjust the analysis no matter what.

TAKEAWAYS

Three things to walk away with.

1 Design controls who is in your trial.

Pocock balances. RPW favors winners. Simple does neither, and that's a choice too.

2 Analysis controls what your estimate means.

Adjustment is non-negotiable in logistic regression because of non-collapsibility, even when the design is balanced.

3 Design and analysis interact at small N.

Pocock buys ~10 points of power at $N = 150$ — and zero by $N = 200$. By then, the analysis dominates.

Q & A

Thank you.

Questions and critiques welcome.